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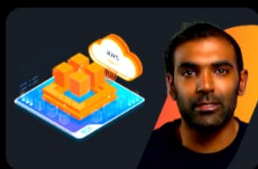
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Courses

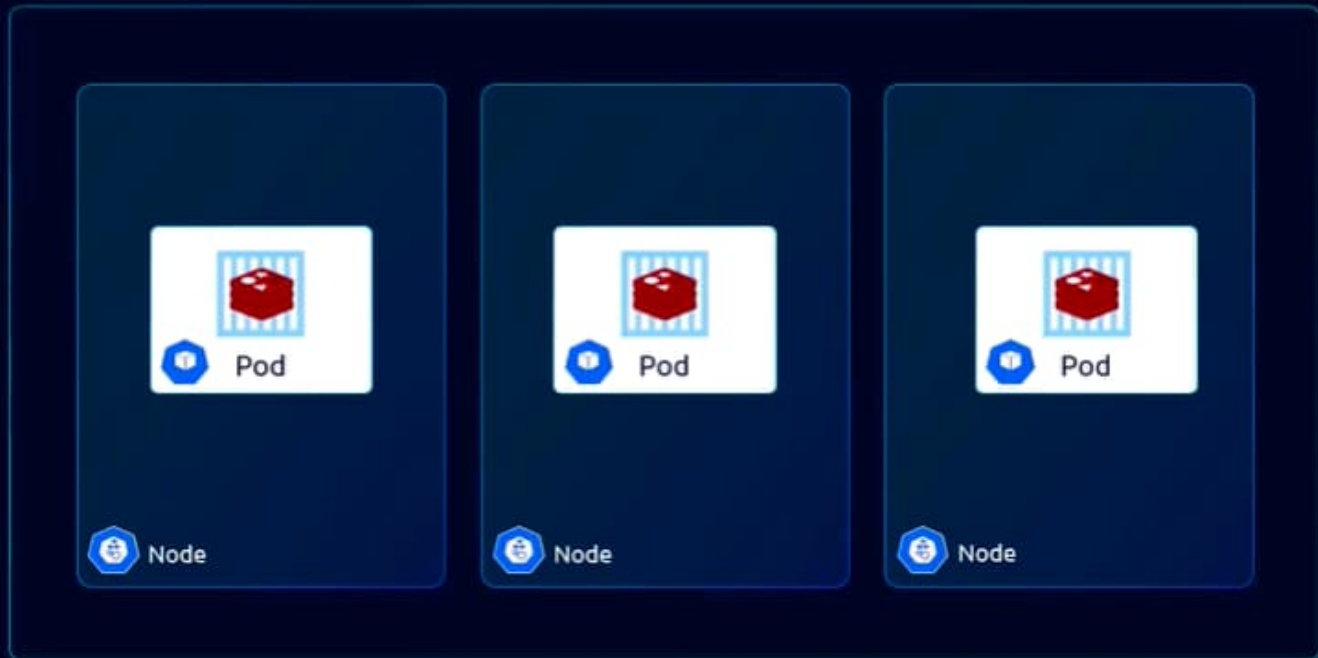


Profile

Pod



Pod



QUESTION 4 OF 4



Create a Pod named `nginx-pod` using the `nginx` image.

Check

Next →



Is the pod running?

||

```
controlplane ~ → kubectl run nginx-pod --image=nginx
pod/nginx-pod created
```

```
controlplane ~ → kubectl get pod
NAME          READY   STATUS    RESTARTS   AGE
nginx-pod     1/1     Running   0           9s
```

```
controlplane ~ →
```

QUESTION 3 OF 4



Preview only — answers can't be changed or retried here.

The Kubernetes control plane URL can be found by running the command `kubectl cluster-info`.

`https://127.0.0.1:6443`



`http://localhost:8080`

`https://0.0.0.0:6443`

`http://127.0.0.1:6443`

||

```
controlplane ~ → kubectl cluster-info
```

```
Kubernetes control plane is running at https://127.0.0.1:6443
```

```
CoreDNS is running at https://127.0.0.1:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy
```

```
Metrics-server is running at https://127.0.0.1:6443/api/v1/namespaces/kube-system/services/https:metrics-server:https/proxy
```

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

```
controlplane ~ →
```

QUESTION 2 OF 4



What is the version of Kubernetes running on the nodes ?

v1.22.3

v1.19.0

v2.18.2

v1.20.2

v1.35.0+k3s3



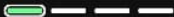
```
controlplane ~ ➔ kubectl get pods
No resources found in default namespace.
```

```
controlplane ~ ➔ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
controlplane	Ready	control-plane	19m	v1.35.0+k3s3

```
controlplane ~ ➔
```

QUESTION 1 OF 4



How many nodes are part of the cluster?

6

5

1

4

3

||

```
controlplane ~ → kubectl get pods
No resources found in default namespace.
```

```
controlplane ~ → kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
controlplane  Ready    control-plane  19m   v1.35.0+k3s3
```

```
controlplane ~ →
```

Container Orchestration



Kubernetes



A container orchestration technology that manages and deploys thousands of containers in a cluster

Container Orchestration



Orchestration



Container Orchestration

Docker Swarm



Kubernetes



MESOS



Kubernetes Basics

13 Lessons

84%



Course Introduction

⌚ 00:52



Kubernetes Overview & Kubectl

⌚ 06:02



Lab: Kubectl



Pods

⌚ 04:31



Lab: Pods



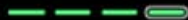
YAML - Part 1

⌚ 03:29



Lab: YAML - Part 1





We have updated the `/home/bob/playbooks/practice.yaml` file with some data for `apple`, `orange` and `mango`. Just like `apple`, we would like to add additional details for each item like color, weight, etc. Modify the remaining items to match the below data.

orange

color	weight
orange	90g

mango

color	weight
yellow	150g

Check



Next →

```
[bob@student-node playbooks]$ ls
practice.yaml
[bob@student-node playbooks]$ vim practice.yaml
[bob@student-node playbooks]$ cat practice.yaml
- name: apple
  color: red
  weight: 100g
- name: orange
  color: orange
  weight: 90g
- name: mango
  color: yellow
  weight: 150g
[bob@student-node playbooks]$
```

QUESTION 3 OF 4

We have updated the `/home/bob/playbooks/practice.yaml` file with a dictionary named `employee`. Add the remaining properties to it using information from the table below.

Key/Property	Value
name	john
gender	male
age	24

Check

Next →

```
[bob@student-node playbooks]$ ls
practice.yaml
[bob@student-node playbooks]$ cat practice.yaml
employee:
  name: john
[bob@student-node playbooks]$ vim practice.yaml
[bob@student-node playbooks]$ cat practice.yaml
employee:
  name: john
  gender: male
  age: 24
[bob@student-node playbooks]$
```

QUESTION 2 OF 4



We have updated the `/home/bob/playbooks/practice.yaml` file with an array of apples. Add a new apple to the list to make it a total of `4`.

Check



Next →



Is a new apple added to the list to make it a total of 4 in "practice.yaml"?

```
[bob@student-node playbooks]$ ls
practice.yaml
[bob@student-node playbooks]$ vim practice.yaml
[bob@student-node playbooks]$ cat practice.yaml
- apple
- apple
- apple
- apple
[bob@student-node playbooks]$
```

QUESTION 1 OF 4



We have updated the `/home/bob/playbooks/practice.yaml` file with the key `name` and value `apple`. Add some additional properties (given below) to the dictionary.

name= `apple`

color= `red`

weight= `90g`

Check



Next →



Are some additional properties added to the dictionary in "practice.yaml" file?

```
ls
[ bob@student-node ~ ]$ ls
playbooks
[ bob@student-node ~ ]$ cd playbooks/
[ bob@student-node playbooks ]$ ls
practice.yaml
[ bob@student-node playbooks ]$ vim practice.yaml
[ bob@student-node playbooks ]$ cat practice.yaml
name: apple
color: red
weight: 90g
[ bob@student-node playbooks ]$
```

||

YAML – Advanced

Key-Value/Dictionary/Lists

```
Fruits:
```

```
  - Banana:
```

```
    Calories: 105
```

```
    Fat: 0.4 g
```

```
    Carbs: 27 g
```

```
  - Grape:
```

```
    Calories: 62
```

```
    Fat: 0.3 g
```

```
    Carbs: 16 g
```


YAML

Key-Value Pair

```
Fruit: Apple
Vegetable: Carrot
Liquid: Water
Meat: Chicken
```

Array/Lists

```
Fruits:
- Orange
- Apple
- Banana

Vegetables:
- Carrot
- Cauliflower
- Tomato
```

Dictionary/Map

```
Banana:
  Calories: 105
  Fat: 0.4 g
  Carbs: 27 g

Grapes:
  Calories: 62
  Fat: 0.3 g
  Carbs: 16 g
```

YAML

XML

JSON

YAML

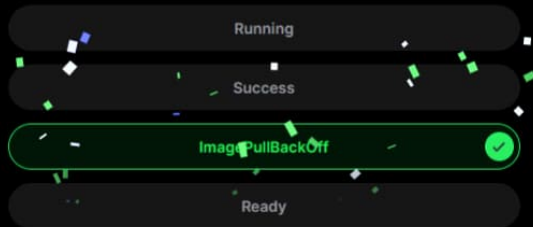
```
<Servers>
  <Server>
    <name>Server1</name>
    <owner>John</owner>
    <created>12232012</created>
    <status>active</status>
  </Server>
</Servers>
```

```
{
  Servers: [
    {
      name : Server1,
      owner : John,
      created : 12232012,
      status : active,
    }
  ]
}
```

```
Servers:
  - name: Server1
    owner: John
    created: 12232012
    status: active
```

What is the state of the pod `webapp` ?

Wait for it to finish the `ContainerCreating` state.



`controlplane ~` → `kubectl get pods -o wide`

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE	READ
INESS GATES									
newpods-j6brc	1/1	Running	0	3m36s	10.22.0.9	controlplane	<none>		<non
e>									
newpods-pj8w8	1/1	Running	0	3m36s	10.22.0.10	controlplane	<none>		<non
e>									
newpods-v6vtx	1/1	Running	0	3m36s	10.22.0.11	controlplane	<none>		<non
e>									
nginx	1/1	Running	0	2m47s	10.22.0.12	controlplane	<none>		<non
e>									
webapp	1/2	ErrImagePull	0	35s	10.22.0.13	controlplane	<none>		<non
e>									

||

`controlplane ~` → `kubectl get pods -o wide`

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE
READINESS GATES								
newpods-j6brc	1/1	Running	0	3m54s	10.22.0.9	controlplane	<none>	
<none>								
newpods-pj8w8	1/1	Running	0	3m54s	10.22.0.10	controlplane	<none>	
<none>								
newpods-v6vtx	1/1	Running	0	3m54s	10.22.0.11	controlplane	<none>	
<none>								
nginx	1/1	Running	0	3m5s	10.22.0.12	controlplane	<none>	
<none>								
webapp	1/2	ImagePullBackOff	0	53s	10.22.0.13	controlplane	<none>	
<none>								

QUESTION 4 OF 5

Preview only — answers can't be changed or retried here.

Which nodes are these pods placed on?

master

controlplane

controlplane & node01

node02

```
controlplane ~ → kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
newpods-j6brc 1/1     Running   0           2m22s
newpods-pj8w8 1/1     Running   0           2m22s
newpods-v6vtx 1/1     Running   0           2m22s
nginx         1/1     Running   0           93s
```

```
controlplane ~ → kubectl get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP           NODE          NOMINATED NODE   READINESS GATES
newpods-j6brc 1/1     Running   0           2m51s  10.22.0.9    controlplane  <none>           <none>
newpods-pj8w8 1/1     Running   0           2m51s  10.22.0.10   controlplane  <none>           <none>
newpods-v6vtx 1/1     Running   0           2m51s  10.22.0.11   controlplane  <none>           <none>
nginx         1/1     Running   0           2m2s   10.22.0.12   controlplane  <none>           <none>
```

```
controlplane ~ →
```

QUESTION 3 OF 5

Preview only — answers can't be changed or retried here.

How many pods are created now?

Note: We have created a few more pods. So please check again in the current(default) namespace.

4



0

7

5

```
controlplane ~ → kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
newpods-j6brc	1/1	Running	0	79s
newpods-pj8w8	1/1	Running	0	79s
newpods-v6vtx	1/1	Running	0	79s
nginx	1/1	Running	0	30s

```
controlplane ~ → kubectl get pods |tail -n +2 | wc -l  
4
```

```
controlplane ~ →
```

QUESTION 2 OF 5

Create a new pod with the `nginx` image. 

Check 

Next 

 Image name: `nginx`

`controlplane ~`  `kubectl get pods`

NAME	READY	STATUS	RESTARTS	AGE
<code>newpods-j6brc</code>	1/1	Running	0	10s
<code>newpods-pj8w8</code>	1/1	Running	0	10s
<code>newpods-v6vtx</code>	1/1	Running	0	10s

`controlplane ~`  `kubectl run --image nginx`

error: NAME is required for run

See 'kubectl run -h' for help and examples

`controlplane ~`  `kubectl run nginx --image nginx`
pod/nginx created

||

`controlplane ~`  `kubectl get pods`

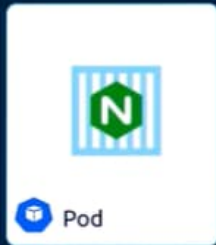
NAME	READY	STATUS	RESTARTS	AGE
<code>newpods-j6brc</code>	1/1	Running	0	54s
<code>newpods-pj8w8</code>	1/1	Running	0	54s
<code>newpods-v6vtx</code>	1/1	Running	0	54s
<code>nginx</code>	0/1	ContainerCreating	0	5s

`controlplane ~` 

Pods Again!

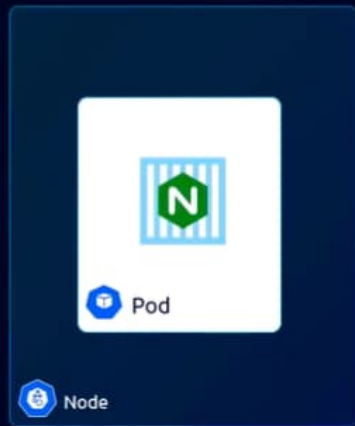
```
$ kubectl run nginx --image nginx
```

```
$ kubectl get pods
```

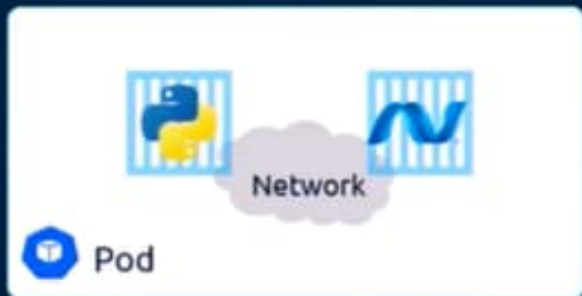


Pods Again!

```
terminal  
$ kubectl run nginx --image nginx
```



Multi-Container Pods

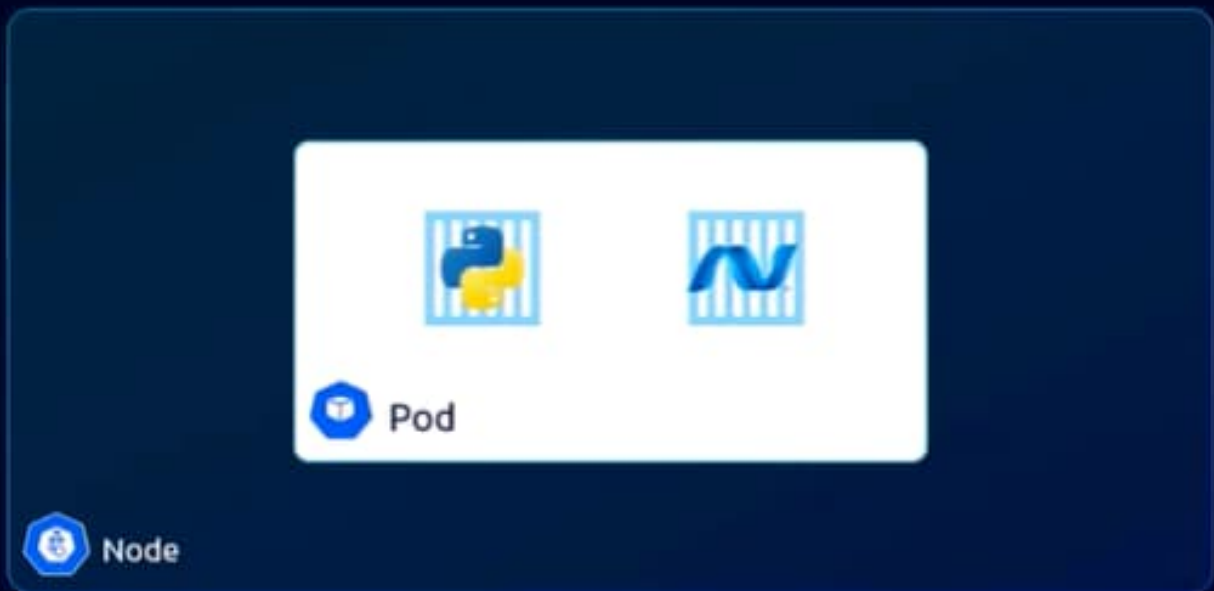


Helper
Containers



Node

Multi-Container Pods



Pod



Pod



Pod

